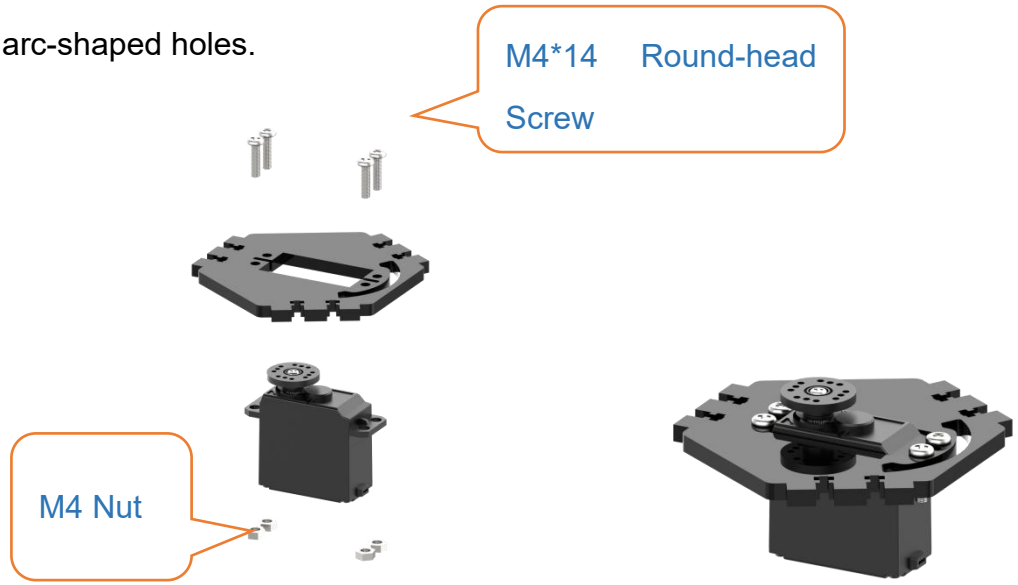
# 1.2 Wiring and Charging Instruction

## 1. Assembly and Wiring

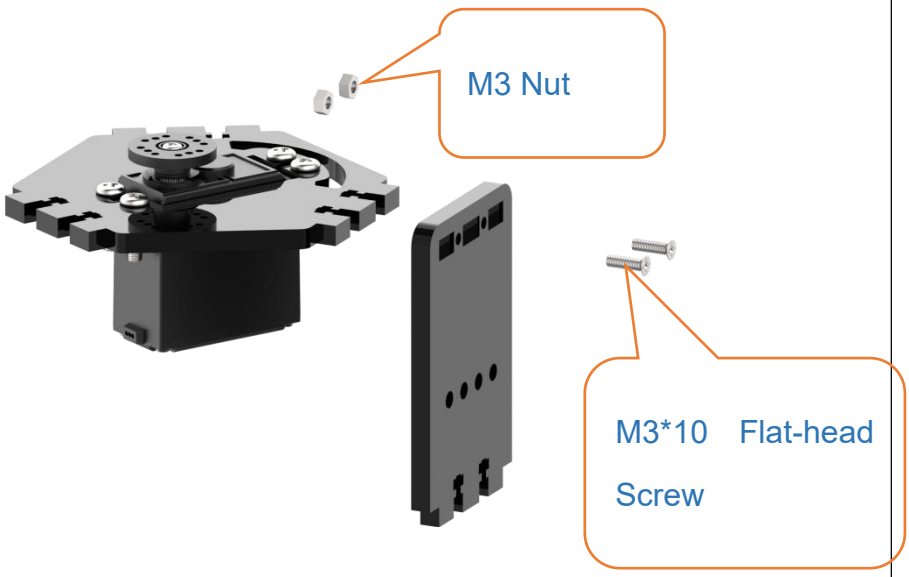
This section takes the uHand UNO starter kit as an example for installation, and the installation method is also suitable for other versions.

1) Install the servo to the servo board with 4 M4*14 round-head screws and 4 M4 nuts.

Please ensure that the servo horn is kept away from the side with the arc-shaped holes.



2) Install the square board to the servo board with 2 M3*10 flat-head screws and 2 M3 nuts.

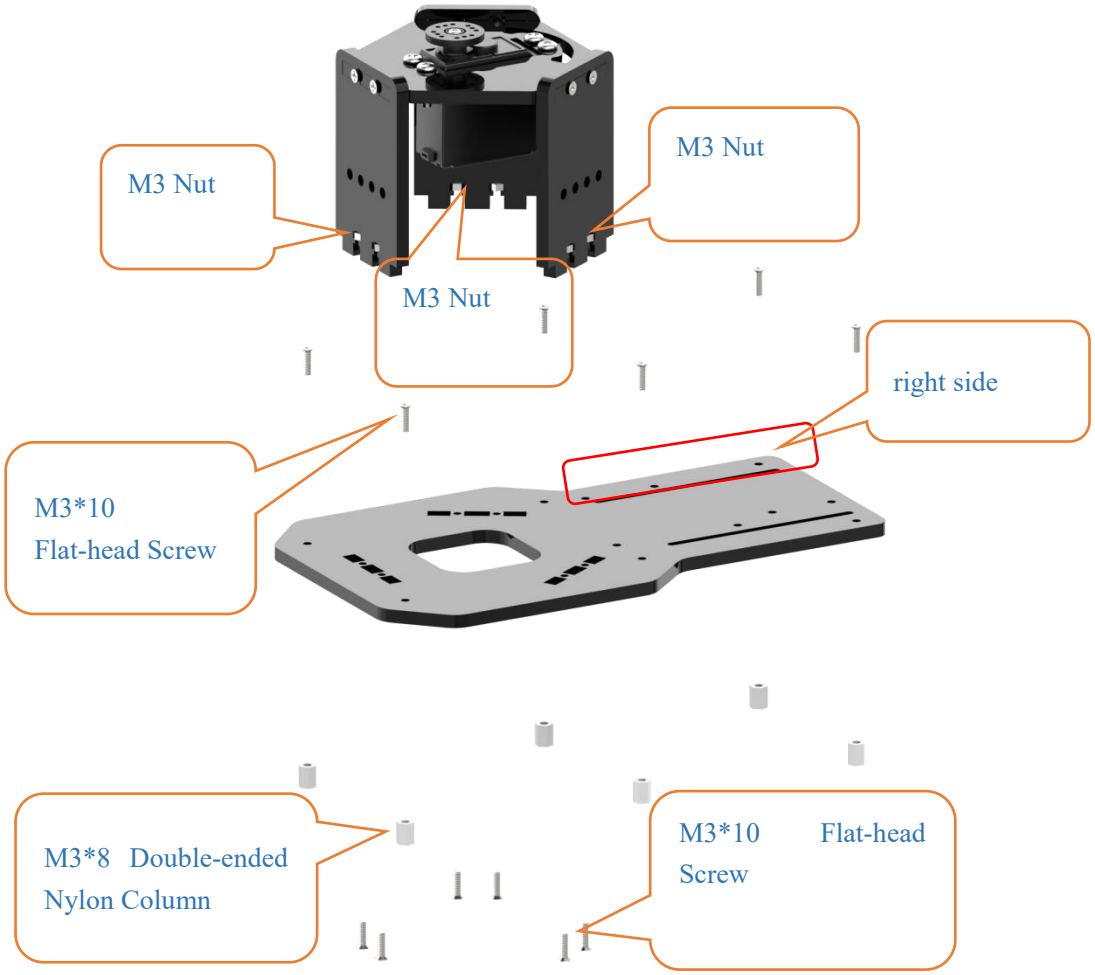


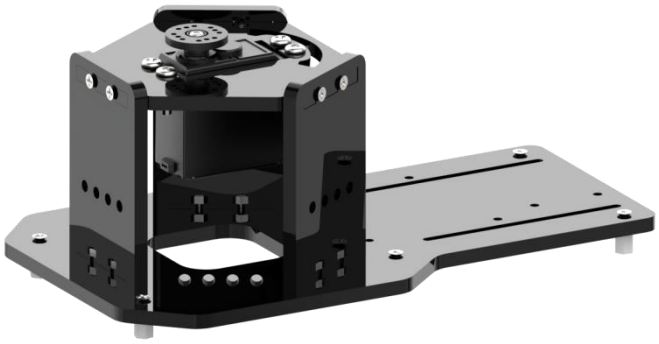

Install the other two square boards in the same way, and the installation is completed as follows:



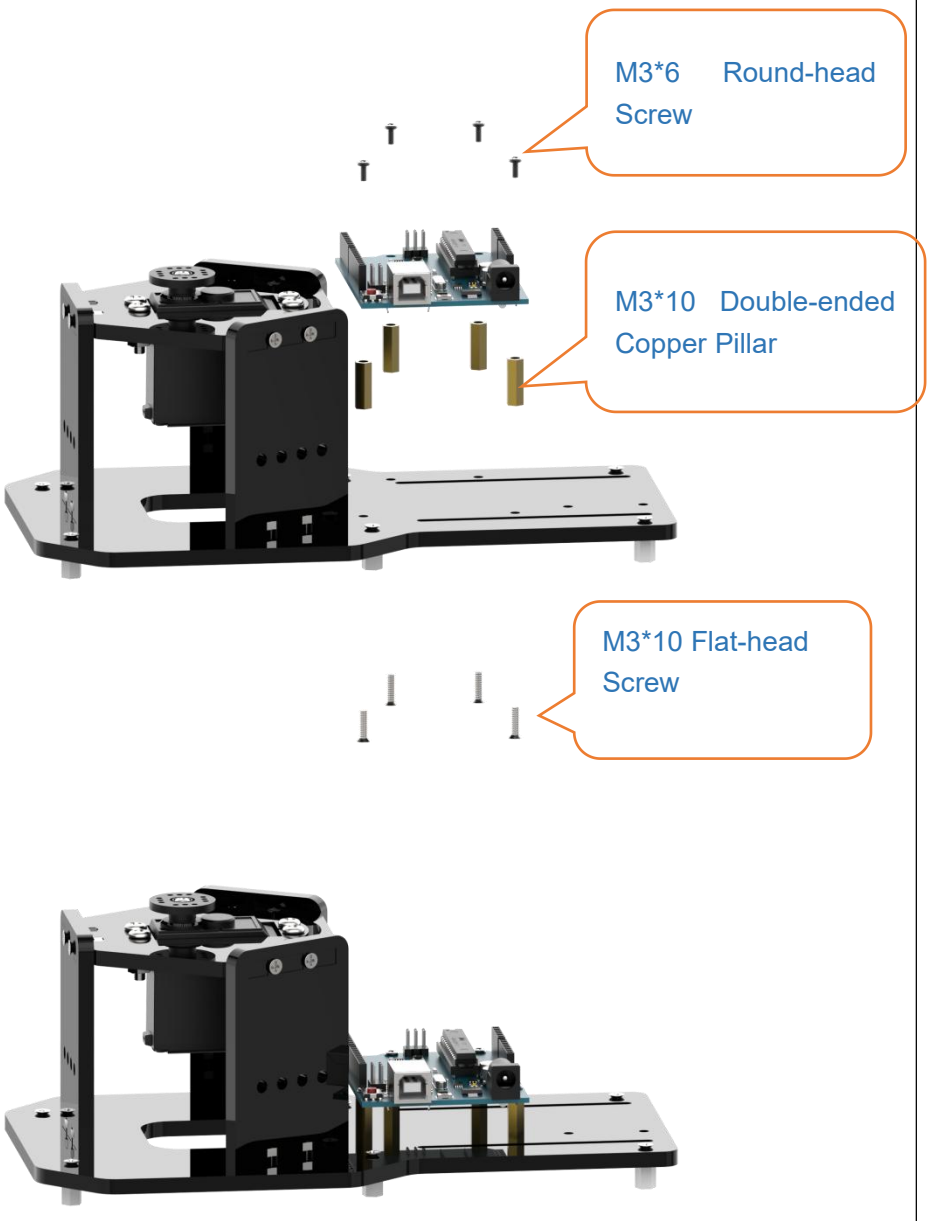
3) Fix the square boards on the baseboard with 6 M3*10 flat-head screws and 6 M3 nuts, and fix beneath the baseboard with 6 M3*10 flat-head screws and 6 M3*8 double-ended nylon columns.

Note: The side of the baseboard with fewer holes should be located on the right side of the uHand UNO robotic hand (with the robotic hand as the first view).

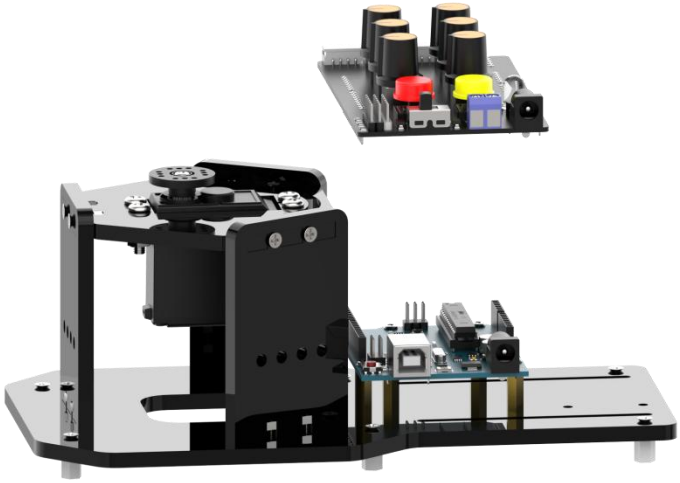


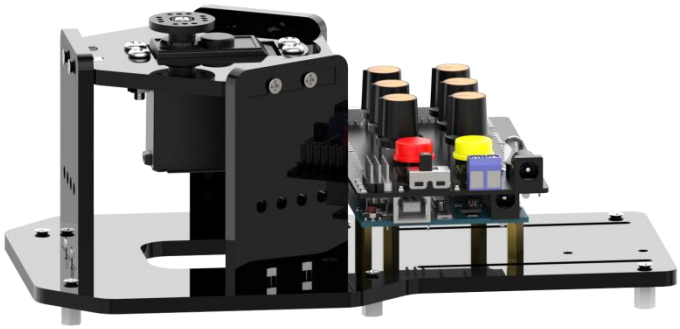


4) Fix 4 M3*10 double-ended copper pillars on Arduino UNO with 4 M3*6 round-head screws, then fix the Arduino UNO on the baseboard with 4 M3*10 flat-head screws.



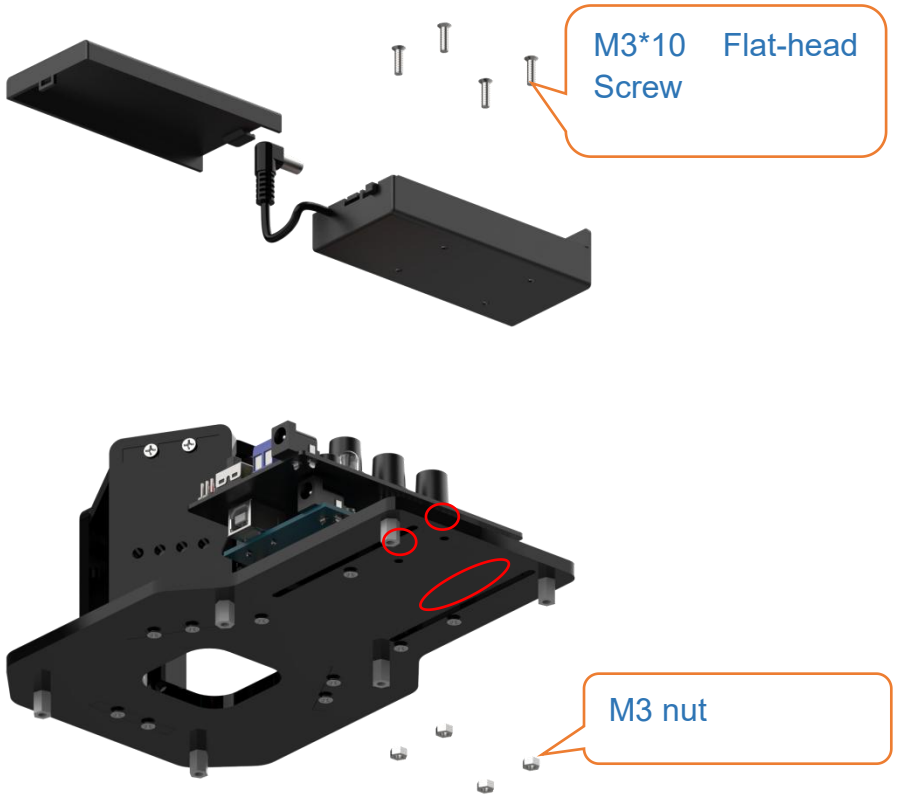
5) Align the pins on the expansion board with the corresponding pins on the Arduino UNO.





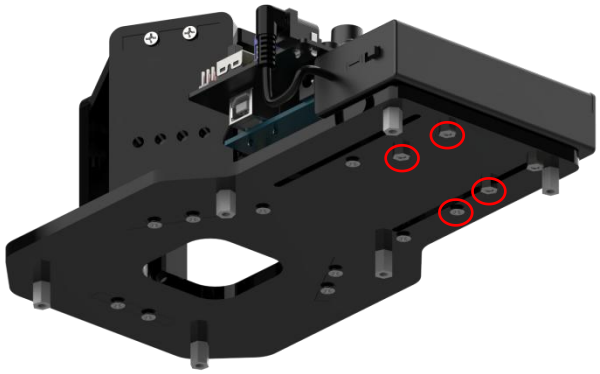
6) Install the battery box on the baseboard with 4 M3*10 flat-head screws and 4 M3 nuts.

Note: There are two holes installed in the gap of the baseboard.

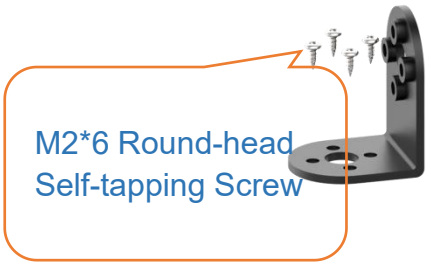



Please switch the battery box to “OFF” to put the batteries into it, and connect the power wire to the DC port on the expansion board.



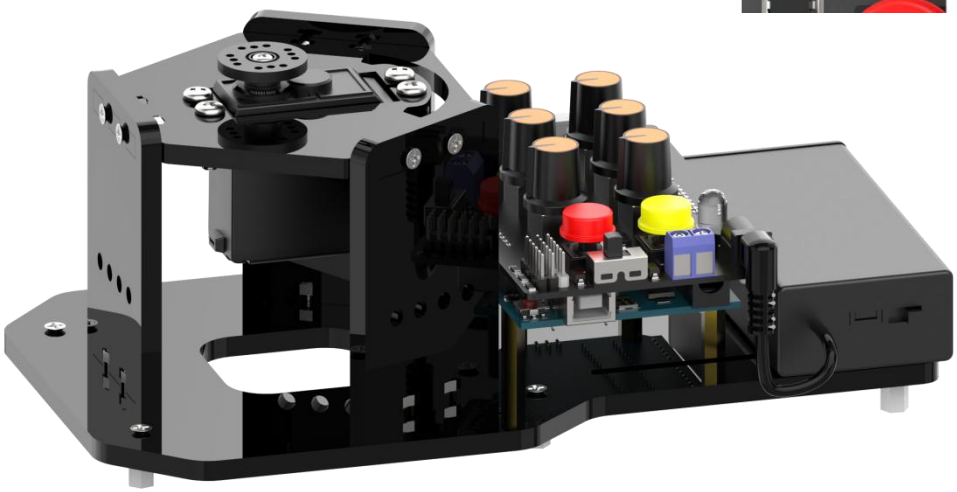


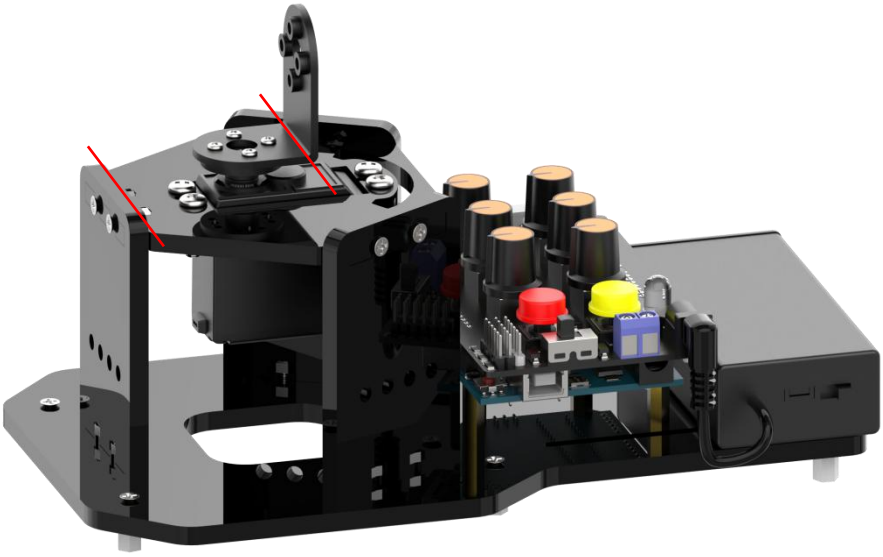
7) Connect the pan-tilt servo to a servo port (with the white wire connected to S (the signal wire), the red wire connected to the positive pole, and the black wire connected to the negative pole), and power the uHand UNO to enter the neutral position. After that, install the L-shaped bracket on the servo horn with 4 M2*6 round-head self-tapping screws (please ensure the bottom of the L-shaped bracket parallels the baseboard).



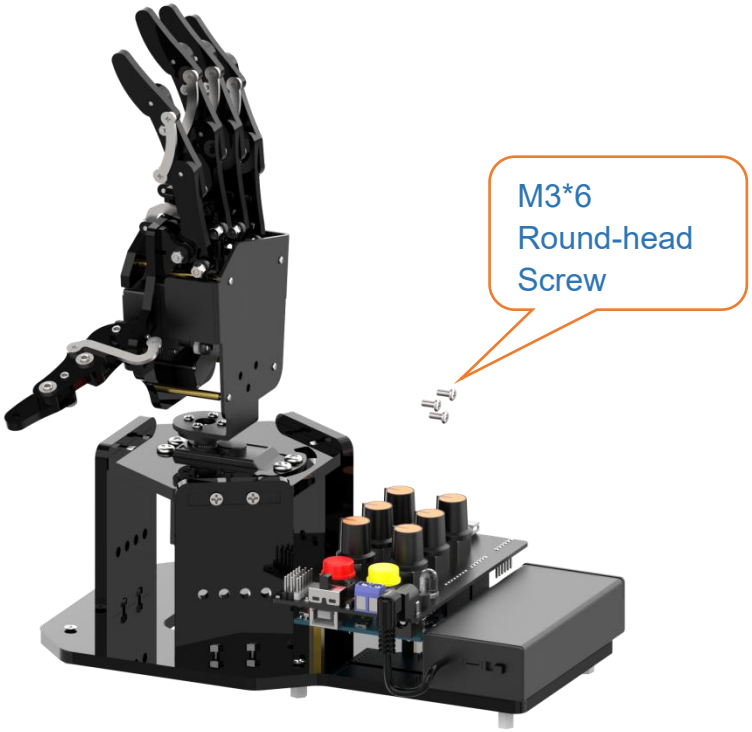








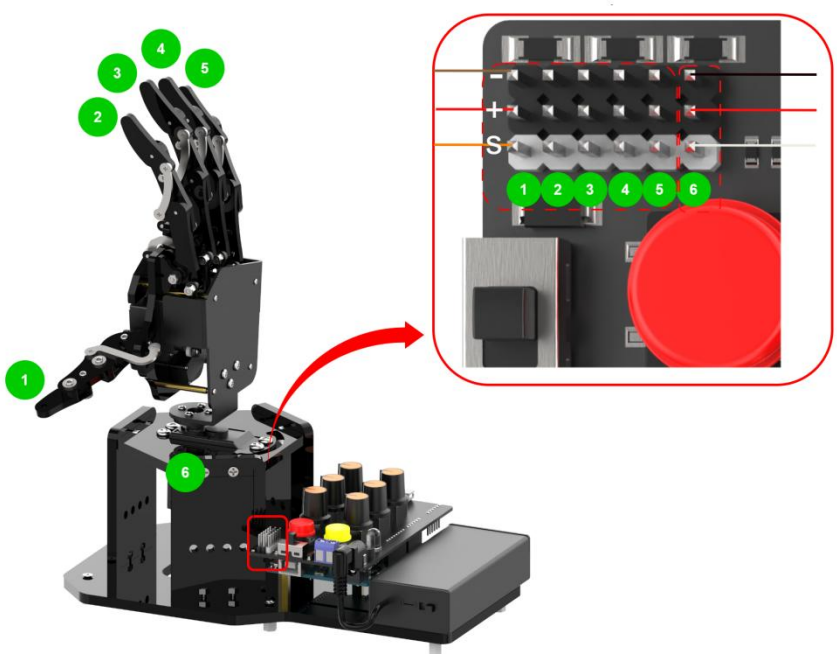
8) Install the robotic hand to the L-shaped bracket with 3 M3*6 round head screws.

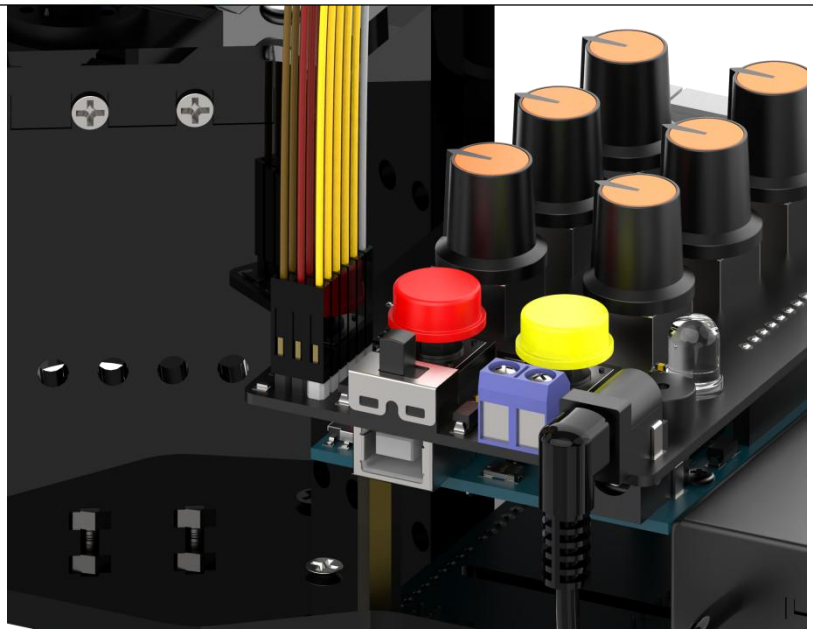




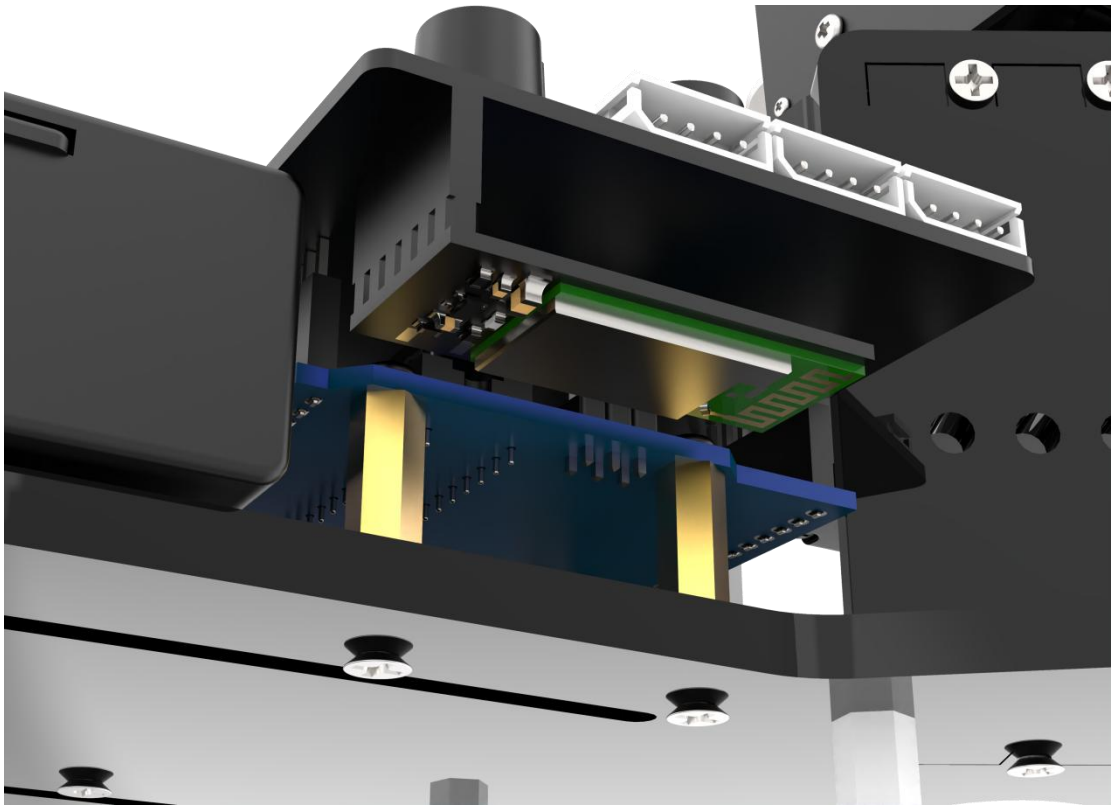

9) Connect servo wires to the corresponding servo ports one by one according to the ID numbers on the labels.

! Please ensure the positive and negative poles of the servos are connected correctly. (No.1 to No.1 ports: connect orange wire with S (signal wire), red wire with positive pole, and brown wire with negative pole; No.6 port: connect white wire with S (signal wire), red wire with positive pole (+), and black wire with negative pole (-).)





10) Align the pins of the Bluetooth module with the pins on the expansion board.



## 2. Charging Instruction

1) Place two 18650 batteries into the battery charger according to positive and negative polarity (positive to positive and negative to negative).




2) Connect the matching USB cable to the power supply first, then plug it

into the USB port of the charger for charging.

Note: You need to prepare a charger head with 5V 3A.




3) When charging, the indicator light will be red. When the batteries are fully charged, the indicator turns green. Please unplug the charger in time to avoid overcharging.

## 3. Battery Usage Guideline

Due to transplantation restrictions, lithium batteries cannot be fully charged during transit. Please fully charge them for your first usage.

1) Please use the battery charger and USB cable provided with the uHand UNO to charge the batteries.

2) The power indicator will turn red during charging, and it will turn green when the batteries are fully charged. Please disconnect the power supply in time to avoid overcharging the batteries.

3) If the robotic hand will not be used for an extended period, please fully charge the batteries, switch the battery box to “OFF”, and unplug the power

wire of the battery box. Store it in a cool and dry place.

4) Official Statement: Our company hereby declares that we shall not be held responsible for any damages to the product, financial losses, or safety accidents caused by failure to comply with the “Precaution and Battery Usage Guideline” of this manual.

## 4. Precaution

1) Please ensure the correct connection of the servo and power wires.

2) Do not forcefully move the servo after powering on to avoid servo damage.

3) Do not keep the servo in a limited position for an extended period to avoid servo blocking.

4) Please ensure that the GPS location and Bluetooth of your phone are turned on when connecting to the app.

5) Please switch the battery box to “OFF” when connecting the Type-B cable. This action prevents the cable from accidentally touching the power pins on the expansion board, which may cause a short circuit.

6) Remove the Bluetooth module while downloading the program to avoid program download failure due to serial conflicts.